

EE 172 – UNIT 4

Transformers

Short Detailed Notes

4.1 INTRODUCTION

What is a Transformer?

- A **stationary AC voltage-changing device**
- Transfers electrical energy from one circuit to another
- **Same frequency** on both sides
- Can **step up** or **step down** voltage

Why Transformers are Important

- Power generated at **low/medium voltages** (e.g., 13.8 kV)
- **Stepped up** for transmission (e.g., 161 kV)

Purpose of high voltage transmission:

- 1 **Reduce power losses** (I^2R losses)
- 2 **Reduce voltage drops**

Key Characteristics

Feature	Description
Moving parts	None (static device)
Construction	Simple, rugged, durable
Maintenance	Very little required
Efficiency	Very high (up to 99%)
Operation	Electromagnetic induction

4.2 PRINCIPLE OF OPERATION

Based On

- **Faraday's Law** of Electromagnetic Induction
- **Lenz's Law**

Basic Construction

- **Primary winding (P)** – connected to AC supply
- **Secondary winding (S)** – connected to load
- **Common laminated iron core**
- Windings insulated from each other

How It Works

- 1 **AC applied to primary** → Alternating current flows
- 2 **Alternating current** → Produces **alternating magnetic flux** (mutual flux)
- 3 **Mutual flux** links both primary and secondary windings
- 4 **Changing flux** → Induces emf in secondary winding
- 5 **Secondary circuit closed** → Secondary current flows

Important Points

- The secondary current also contributes to the magnetic field
- Total magnetic field = contributions from **both** primary and secondary currents

4.3 BASIC RELATIONSHIPS

Assumptions

- 1 Source voltage is sinusoidal → Induced emf is sinusoidal
- 2 **No flux leakage** – all flux links both coils

Definitions

Symbol	Meaning
N_1	Number of turns on primary
N_2	Number of turns on secondary
ϕ_{max}	Maximum flux (Webers)
f	Frequency (Hz)
E_1	Primary induced voltage
E_2	Secondary induced voltage

Flux Equation

$$\phi = \phi_{\text{max}} \times \sin(\omega t)$$

Induced EMF Equations

Instantaneous secondary emf:

$$e_2 = -d(N_2\phi)/dt = -\phi_{\max} \times \omega \times N_2 \times \cos(\omega t)$$

(Negative sign = Lenz's Law – opposes the flux that induces it)

Maximum value:

$$E_{2\max} = 2\pi \times f \times \phi_{\max} \times N_2$$

RMS value (fundamental equation):

$$E_2 = 4.44 \times f \times \phi_{\max} \times N_2$$

$$E_2 = 4.44 \times f \times B_{\max} \times A \times N_2$$

Primary voltage:

$$E_1 = 4.44 \times f \times \phi_{\max} \times N_1$$

Voltage Transformation Ratio (Turns Ratio)

$$n = E_1/E_2 = N_1/N_2$$

For Step-Up Transformer: $N_2 > N_1 \rightarrow E_2 > E_1$

For Step-Down Transformer: $N_2 < N_1 \rightarrow E_2 < E_1$

Example 1 – Transformer Calculations

Problem: 250 kVA, 11,000/400 V, 50 Hz, $N_2 = 80$ turns

(a) Primary and secondary currents:

$$I_1 = S/V_1 = 250,000/11,000 = 22.7 \text{ A}$$

$$I_2 = S/V_2 = 250,000/400 = 625 \text{ A}$$

(b) Number of primary turns:

$$N_1 = (E_1/E_2) \times N_2 = (11,000/400) \times 80 = 2,200 \text{ turns}$$

(c) Maximum flux:

$$\phi_{\max} = E_2 / (4.44 \times f \times N_2)$$

$$\phi_{\max} = 400 / (4.44 \times 50 \times 80) = 22.5 \text{ mWb}$$

4.4 OPERATING CONDITIONS

4.4.1 No-Load Condition

What It Means:

- Secondary winding is **open-circuited**
- No current flows in secondary

What Happens:

- Primary acts like a **highly inductive circuit**
- Current is called **no-load current** (I_0)
- Establishes **no-load flux (useful flux)**

Functions of No-Load Flux:

- 1 Induces voltages in **both** primary and secondary
- 2 **Magnetizes the core**
- 3 Causes **core losses** (hysteresis + eddy current)

4.4.2 Load Condition

What Happens When Load is Connected:

- 1 Secondary current (I_2) flows due to induced voltage (E_2)
- 2 Secondary sets up **secondary mmf** ($N_2 I_2$)
- 3 Secondary flux **opposes** the primary (useful) flux
- 4 Primary takes **additional current** (I_1') to maintain flux

Mmf Balance Equation:

$$N_1 \times I_1' = N_2 \times I_2$$

Total Primary Current:

$$I_1 = I_0 + I_1' \text{ (phasor sum)}$$

Current Transformation Ratio:

$$I_1'/I_2 = N_2/N_1$$

Comparing with Voltage Ratio:

$$E_1/E_2 = N_1/N_2 = I_2/I_1'$$

Key Insight: Current transformation ratio is the **inverse** of voltage transformation ratio!

4.5 EQUIVALENT CIRCUIT

Why We Need Equivalent Circuits

Actual transformer has:

- **Resistance** in windings
- **Leakage reactance** (flux that doesn't link both coils)

Equivalent circuit models these imperfections

Equivalent Circuit Components

Component	Description
R_1	Primary winding resistance
X_1	Primary leakage reactance
R_2	Secondary winding resistance
X_2	Secondary leakage reactance
I_1	Full-load primary current
I_2	Full-load secondary current

Referred Quantities

Why Refer to One Side?

- Simplifies calculations
- Can combine parameters

Referred to Primary:

$$\begin{aligned}R_2' &= (N_1/N_2)^2 \times R_2 \\X_2' &= (N_1/N_2)^2 \times X_2 \\Z_2' &= (N_1/N_2)^2 \times Z_2\end{aligned}$$

Total Equivalent Values (Referred to Primary)

$$\begin{aligned}R_{e1} &= R_1 + R_2' = R_1 + (N_1/N_2)^2 \times R_2 \\X_{e1} &= X_1 + X_2' = X_1 + (N_1/N_2)^2 \times X_2 \\Z_{e1} &= \sqrt{(R_{e1})^2 + (X_{e1})^2}\end{aligned}$$

4.6 VOLTAGE REGULATION

Definition

The change in secondary terminal voltage from no-load to full-load, expressed as a percentage of no-load voltage.

Formula

$$VR = (E_2 - V_2)/E_2$$

Where: E_2 = No-load secondary voltage, V_2 = Full-load secondary voltage

In Terms of Referred Quantities

$$VR = I_1(R_{e1}\cos\phi \pm X_{e1}\sin\phi)/V_1$$

- + sign → Lagging power factor
- - sign → Leading power factor

Secondary Terminal Voltage

$$V_2 = E_2(1 - VR) \text{ (For lagging pf)}$$

$$V_2 = E_2(1 + VR) \text{ (For leading pf)}$$

Key Points

- **Lagging pf** → Voltage drops (negative regulation)
- **Leading pf** → Voltage rises (positive regulation)
- Good regulation means **small change** from no-load to full-load

Example 2 – Voltage Regulation

Problem: 100 kVA, 400/80 turns, $R_1=0.3\Omega$, $R_2=0.01\Omega$, $X_1=1.1\Omega$, $X_2=0.035\Omega$, $V_1=2200V$

(a) Equivalent impedance referred to primary:

$$\begin{aligned}R_{e_1} &= 0.3 + (400/80)^2 \times 0.01 = 0.55\Omega \\X_{e_1} &= 1.1 + (400/80)^2 \times 0.035 = 1.975\Omega \\Z_{e_1} &= \sqrt{(0.55^2 + 1.975^2)} = 2.05\Omega\end{aligned}$$

(b)(i) Voltage regulation at 0.8 lagging:

$$\begin{aligned}I_1 &= S/V_1 = 100,000/2200 = 45.45A \\ \cos\phi &= 0.8 \rightarrow \sin\phi = 0.6 \\ VR &= I_1(R_{e_1}\cos\phi + X_{e_1}\sin\phi)/V_1 \\ VR &= 45.45(0.55 \times 0.8 + 1.975 \times 0.6)/2200 \\ VR &= 0.0336 = 3.36\%\end{aligned}$$

$$\begin{aligned}E_2 &= 2200 \times (80/400) = 440V \\ V_2 &= 440(1 - 0.0336) = 425.2V\end{aligned}$$

(b)(ii) Voltage regulation at 0.8 leading:

$$\begin{aligned}VR &= 45.45(0.55 \times 0.8 - 1.975 \times 0.6)/2200 \\ VR &= -0.0154 = -1.54\% \\ V_2 &= 440(1 + 0.0154) = 447V\end{aligned}$$

Key Insight: Leading power factor causes **voltage rise** (positive regulation)!

4.7 TRANSFORMER LOSSES

Two Main Types of Losses

Type	Also Called	Cause
Core (Iron) Losses	Iron losses	Alternating flux in core
Copper Losses	I^2R losses	Current in windings

4.7.1 Iron (Core) Losses

Two Components:

(a) Hysteresis Loss

- Due to **magnetization and demagnetization** of core
- Each cycle of AC causes **molecular friction**

Steinmetz Formula:

$$P_{\text{hys}} = k_h \times V_c \times f \times B_{\text{max}}^{1.6}$$

Where: k_h = Hysteresis constant, V_c = Volume of core, f = Frequency, B_{max} = Maximum flux density

(b) Eddy Current Loss

- Due to **circulating currents** induced in core
- Laminations **reduce** eddy currents

$$P_{\text{eddy}} = (k_{\text{eddy}} \times A^2 \times f^2 \times B_m^2) / \rho$$

Where: A = Cross-sectional area, ρ = Resistivity of material

Reducing Iron Losses:

- 1 **Silicon steel** laminations (reduces hysteresis)
- 2 **Thin laminations** (reduces eddy currents)
- 3 Laminations **insulated** from each other

4.7.2 Copper (I^2R) Losses

- Due to **resistance** of windings
- **Proportional to square of current**

$$P_{\text{cu}} = I_1^2 R_1 + I_2^2 R_2$$

Reducing Copper Losses:

- Use **larger conductors**
- Use **better conducting material** (copper)

4.8 TRANSFORMER TESTS

Why Tests are Performed

- Determine **voltage regulation (VR)**
- Determine **efficiency (η)**
- Find **equivalent circuit parameters**
- **Without actually loading** the transformer

4.8.1 Open-Circuit (No-Load) Test

Connection:

- Secondary: **Open**
- Primary: Connected to rated voltage via VARIAC

Measurements:

Instrument	Measures
Voltmeter	Rated voltage
Ammeter	No-load current ($\sim 2.5\%$ of rated)
Wattmeter	Core losses (iron losses)

What We Get:

- Core/Iron losses (P_{fe})
- Magnetizing reactance (X_m)
- Core loss resistance (R_c)

4.8.2 Short-Circuit Test

Connection:

- Secondary: **Shorted** through ammeter
- Primary: Connected to **reduced voltage** via VARIAC

Procedure: Increase voltage until **rated/full-load current** flows

Measurements:

Instrument	Measures
Voltmeter	Short-circuit voltage
Ammeter	Full-load current
Wattmeter	Copper losses

What We Get:

- Copper losses (P_{cu}) at full load

- Equivalent resistance (R_e)
- Equivalent reactance (X_e)

Important: S/C voltage is usually **small** (5–10% of rated voltage)

4.9 EFFICIENCY

Definition

$$\eta = \text{Output Power} / \text{Input Power}$$

Alternate Forms

$$\begin{aligned}\eta &= \text{Output} / (\text{Output} + \text{Losses}) \\ \eta &= (\text{Input} - \text{Losses}) / \text{Input} \\ \eta &= 1 - \text{Losses}/\text{Input}\end{aligned}$$

Detailed Formula

$$\eta = (V_2 \times I_2 \times \cos\phi_2) / (V_2 \times I_2 \times \cos\phi_2 + P_{fe} + P_{cu})$$

Where: P_{fe} = Iron losses (from O/C test), P_{cu} = Copper losses (from S/C test)

At Full Load

$$\eta = (S_n \times \text{pf}) / (S_n \times \text{pf} + P_{fe} + P_{cu,fl})$$

Key Points

- **Iron losses** are **constant** (fixed) – independent of load
- **Copper losses** are **variable** – proportional to load²
- Maximum efficiency occurs when:

$$P_{cu} = P_{fe}$$

(i.e., variable losses = fixed losses)

Example 3 – Efficiency Calculation

Problem: 50 kVA transformer – O/C test: $P = 430\text{W}$ (iron loss), S/C test: $P = 525\text{W}$ (copper loss), Load: Full-load, 0.7 lagging pf

$$\begin{aligned}\eta &= (S_n \times \text{pf}) / (S_n \times \text{pf} + P_{fe} + P_{cu}) \\ \eta &= (50,000 \times 0.7) / (50,000 \times 0.7 + 430 + 525) \\ \eta &= 35,000 / 35,955 \\ \eta &= 0.9734 = 97.34\%\end{aligned}$$

4.10 COOLING OF TRANSFORMERS

Why Cooling is Important

- 1 Prevent **deterioration of insulation**
- 2 **Extend transformer life**
- 3 Prevent **overheating** of core and windings

Two Sources of Heating

- 1 **No-load losses** (core/iron losses) – present even with no load
- 2 **Load losses** (copper losses) – increase with load

Cooling Media

Type	Cooling Medium
Dry-type	Air or gas
Oil-immersed	Mineral oil or synthetic liquid

4.10.1 Advantages of Oil Over Air

Advantage	Explanation
Higher specific heat capacity	Absorbs more heat for same temperature rise
Greater heat conductivity	Transfers heat faster
Higher breakdown strength	~6× better than air → Better insulation at high voltages

Small Transformers:

- <5 kVA → Air-cooled
- Heat carried away by surrounding natural air

Large Transformers:

- **Oil-cooled** preferred
- Especially where high voltages are involved

4.10.2 Cooling Designations

IEC Standard Designations:

Letters	Meaning
O	Oil
A	Air

W	Water
N	Natural circulation
F	Forced circulation

Common Cooling Methods:

Designation	Meaning
ONAN	Oil Natural, Air Natural
ONAF	Oil Natural, Air Forced
OFAN	Oil Forced, Air Natural
OFAF	Oil Forced, Air Forced
OFWF	Oil Forced, Water Forced
AN	Air Natural (dry-type)
AF	Air Forced (dry-type)

What Each Means:

Method	Oil Circulation	Cooling Medium
ONAN	Natural (thermal head)	Natural air
ONAF	Natural	Forced air (fans)
OFAN	Forced (pumps)	Natural air
OFAF	Forced	Forced air
OFWF	Forced	Water

Purpose of Coolant:

- 1 **Transports heat** from windings to dissipation surfaces
- 2 **Insulates** primary from secondary
- 3 Protects against **moisture and contamination**

KEY FORMULAS SUMMARY

Voltage and Turns Ratio

$$\begin{aligned} E_1/E_2 &= N_1/N_2 \\ I_1/I_2 &= N_2/N_1 \text{ (ideal transformer)} \end{aligned}$$

EMF Equation

$$\begin{aligned} E &= 4.44 \times f \times \phi_{\max} \times N \\ E &= 4.44 \times f \times B_{\max} \times A \times N \end{aligned}$$

Equivalent Circuit (Referred to Primary)

$$\begin{aligned} R_{e1} &= R_1 + (N_1/N_2)^2 \times R_2 \\ X_{e1} &= X_1 + (N_1/N_2)^2 \times X_2 \\ Z_{e1} &= \sqrt{(R_{e1})^2 + (X_{e1})^2} \end{aligned}$$

Voltage Regulation

$$\begin{aligned} VR &= (E_2 - V_2)/E_2 \\ VR &= I_1(R_{e1} \cos \phi \pm X_{e1} \sin \phi)/V_1 \end{aligned}$$

Efficiency

$$\begin{aligned} \eta &= \text{Output} / (\text{Output} + \text{Losses}) \\ \eta &= (S \times \text{pf}) / (S \times \text{pf} + P_{fe} + P_{cu}) \end{aligned}$$

Maximum Efficiency Condition

$$P_{cu} = P_{fe} \text{ (at some load current)}$$

TEST COMPARISON

Feature	Open-Circuit Test	Short-Circuit Test
Secondary	Open	Short-circuited
Primary voltage	Full rated voltage	Reduced voltage
Primary current	Small (2.5% of rated)	Full rated current
Measures	Core/Iron losses	Copper losses
Determines	R_c, X_m	R_e, X_e, VR

IMPORTANT CONCEPTS TO REMEMBER

- 1 **Transformer is a static device** – no moving parts
- 2 **Frequency is the same** on primary and secondary
- 3 **Voltage ratio $\propto$ turns ratio**

- 4 **Current ratio** $\propto$ **1/turns ratio**
- 5 **Efficiency can be 99%** or higher
- 6 **Iron losses** are **constant** (fixed)
- 7 **Copper losses** are **variable** ($\propto I^2$)
- 8 **Leading pf** $\rightarrow$ **Voltage rise** (negative VR)
- 9 **Lagging pf** $\rightarrow$ **Voltage drop** (positive VR)
- 10 **Maximum efficiency** when $P_{fe} = P_{cu}$